

OCEANEMBED

SIH 2026 — SIH26066

Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature

Research & Technical Team Dossier

"We evaluated the obvious approaches first, identified their limitations for this specific problem, and then designed OceanEmbed around the smallest set of components that can plausibly capture spatial structure, represent the joint surface ocean state, and reconstruct a coherent subsurface temperature profile."

1. Executive Summary

OceanEmbed is our team's proposed solution to Smart India Hackathon (SIH) 2026 Problem Statement 26066. The objective is to reconstruct 3D subsurface ocean temperature profiles using only 2D surface observations collected by satellites.

This document serves as the single source of truth for our 6-member team. It establishes the scientific foundation, evaluates candidate solutions, and explicitly defines our target architecture: a **Masked Multi-Scale U-Net + Latent Ocean Embedding + Depth-Conditioned Profile Decoder**. We prioritize a strict, component-wise ablation strategy, meaning we will validate simple baselines before proving that our advanced architectural components genuinely improve accuracy against independent ARGO float observations.

2. CURRENT REALITY: What We Know vs What Is Still Unproven

✓ **PS understood:** We know the exact region, variables, and depths required. ✓ **GLORYS identified:** The target reanalysis dataset has been sourced. ✓ **Initial dataset audited:** We programmatically confirmed the initial GLORYS download was only a 2D surface slice (unsuitable). ✓ **Native depth structure investigated:** We identified the exact 36 native GLORYS depths needed to span 0–1000m. **Correct 3D GLORYS pilot pending:** A 7-day 3D pilot is blocked by authentication. **Seven surface datasets pending:** Exact URLs and automated downloads are being investigated. **Baseline ML not yet trained:** No models have been experimentally run. **ARGO validation not yet performed:** Colocation and independent metrics are unproven. **Final architecture not yet proven:** The proposed architecture is our target, but must be experimentally validated.

3. Exact SIH26066 Requirements

[Official Source: SIH Specification]

Requirement	SIH Specification
Region	North Indian Ocean
Latitude	5°N–30°N
Longitude	45°E–105°E
Spatial output	0.25° × 0.25°
Temporal output	Daily
Surface Inputs	SST, SSS, SSH/SLA, Surface U/V current, Wind U/V
Target Output	Subsurface temperature
Target Depths	0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000 m
Training target	GLORYS Global Ocean Reanalysis

Independent validation	Gridded ARGO / INCOIS LAS
Metrics	RMSE, correlation, Bias
Demonstration	Bay of Bengal / Arabian Sea

4. What Are We Actually Solving?

OceanEmbed is fundamentally a **high-dimensional spatiotemporal inverse problem**: infer a 15-level vertical ocean temperature profile at every 0.25° daily grid location from a multi-variable surface observation field.

It operates across three dimensions:

- **Horizontal**: 0.25° × 0.25° spatial context.
- **Vertical**: 15 target depths from surface to 1000 m.
- **Temporal**: Daily evolution.

The problem is NOT simply taking 7 numbers and doing a linear regression to 15 numbers. It is mapping a complex spatial field to an underlying physical structure: Spatiotemporal surface field → Latent ocean state → Vertical temperature profile

5. Problem Formulation

The model receives seven surface variables. (Note: "Expected Relevance" below is a Research Hypothesis to be proven experimentally).

Variable	Physical information	Expected relevance
SST	Surface thermal state	High
SSS	Freshwater/haline structure (density driver)	High
SSH/SLA	Dynamic height / thermocline-related information	High
U/V current	Surface transport / mesoscale dynamics	High
Wind U/V	Surface forcing / Ekman pumping	Medium–High

The model predicts: T(0m), T(5m), T(10m), T(20m), T(30m), T(50m), T(75m), T(100m), T(125m), T(150m), T(200m), T(300m), T(500m), T(700m), T(1000m). The output is a continuous **vertical profile**, which must influence our architecture design.

6. First Eliminate the Obvious Approaches

Before proposing a complex custom architecture, we must explicitly evaluate and demote approaches that either lack necessary capabilities or introduce unneeded complexity.

1. **Point-wise MLP & Linear Regression:**
 - Treats every grid cell independently. Cannot naturally exploit neighboring SST/SSH/current structures (like eddies). Useful as baselines, not final architectures.
2. **XGBoost / LightGBM:**
 - Strong for tabular benchmarks. Does not naturally model dense spatial fields or neighborhood structures. Keep as a benchmark, not a preferred architecture.
3. **Plain CNN:**
 - Captures spatial context. Strong candidate baseline, but profile generation/depth structure is not explicitly modeled.

Use as an important baseline.

4. **Plain U-Net:**

- Strong multi-scale spatial representation suited to dense geospatial reconstruction. But by itself, it does not explicitly create a compact latent ocean-state representation or depth-conditioned profile decoder.

5. **ConvLSTM:**

- Useful for temporal dependencies. However, existing research already uses ConvLSTM for satellite-to-subsurface reconstruction. It should NOT automatically be chosen as the starting architecture. Treat temporal modeling as an extension after proving the spatial architecture.

6. **Pure Transformer / ViT:**

- Powerful attention-based modeling, but potentially expensive/data-hungry. Higher implementation complexity makes it sub-optimal for a first-pass student PoC.

7. **Autoencoder Alone:**

- Useful for representation learning, but compression alone does not solve the supervised subsurface reconstruction problem. Its embedding idea is more useful when integrated into the supervised architecture.

8. **GNN:**

- Can model relationships between nodes, but the target is already represented on a regular 0.25° grid. Adds graph-construction and computational complexity without an obvious first-order advantage. Do not choose it as the initial architecture.

9. **Giant Hybrid Architecture ("Architecture Soup"):**

- CNN + Transformer + ConvLSTM + GNN + GAN + physics loss etc. Explicitly rejected. Complexity must earn its place through measurable improvement.

7. Solution Elimination Matrix

Approach	Strength	Main Limitation	Feasibility	Role in Project	Decision
MLP / Linear	Fast, easy	Ignores spatial context / eddies	High	Baseline only	Demote
XGBoost	Strong nonlinear	Fails at spatial field tasks	High	Benchmark only	Demote
Plain CNN	Captures spatial	Treats depths independently	High	Level 1 Baseline	Keep
Plain U-Net	Multi-scale spatial	Ignores vertical profile structure	High	Level 2 Baseline	Keep
ConvLSTM	Spatial + Temporal	High compute, memory intensive	Medium	Advanced extension	Defer
Pure ViT	Global context	Data-hungry, hard to optimize	Low	Reject as base	Demote
GNN	Irregular grids	Unnecessary for 0.25° regular grid	Low	Reject	Reject

Conclusion: We need an architecture with the spatial capabilities of a U-Net, but augmented to explicitly model the multi-variable surface state and decode it into a coherent vertical profile.

8. Existing Research Landscape

[Literature Review Requirement] We must acknowledge peer-reviewed literature. Existing research has already demonstrated:

- ConvLSTM-based global subsurface reconstruction.
- CNN/LightGBM reconstruction to 1000m at 0.25° resolution.
- Convformer and Graph-attention approaches.
- Attention-enhanced U-Net approaches.

- Uncertainty-aware AI reconstruction.

9. Research Gap

Existing work demonstrates that CNN, ConvLSTM, and Transformer variants can solve this problem. Therefore, our contribution is NOT simply "using deep learning".

We frame OceanEmbed as an engineering/research hypothesis: **"A compact, experimentally validated architecture that combines multi-scale spatial encoding, a learned latent representation of the joint surface state, and explicit depth-conditioned profile decoding, while using independent ARGO validation and component-wise ablation to determine whether each added component actually improves reconstruction."**

10. THE SOLUTION WE SHOULD AIM FOR

The core of our proposal is the **Masked Multi-Scale U-Net + Latent Ocean Embedding + Depth-Conditioned Profile Decoder**.

Overall Data Flow: INPUT (7 surface variables + masks) → Data Harmonization → Masked Multi-Scale U-Net → Latent Ocean Embedding → Depth-Conditioned Profile Decoder → 15-depth temperature profile → ARGO Validation .

11. OceanEmbed Architecture Deep Dive

```
graph TD
    A[7 Surface Variables + Validity Masks] --> B[Data Harmonization]
    B --> C[Masked Multi-Scale U-Net]
    C --> D[Latent Ocean Embedding]
    E[Depth Embedding] --> F[Depth-Conditioned Decoder]
    D --> F
    F --> G[15 Temperature Outputs]
    G --> H[ARGO Independent Validation]
```

12. Component-by-Component Architecture Explanation

A. MASKED MULTI-SCALE U-NET

The model receives the seven surface variables together with validity masks.

- **Why masks matter:** Satellite products contain missing/invalid observations (e.g., clouds blocking SST). The model should distinguish "zero value" from "missing value". Missing-data robustness is part of the engineering solution.
- **Why multi-scale:** It captures local gradients/fronts, mesoscale structures/eddies, and larger-scale basin patterns.

B. LATENT OCEAN EMBEDDING

This is the conceptual core of OceanEmbed. The U-Net bottleneck learns a compact latent representation of the joint surface ocean state.

- **The key idea:** Instead of independently mapping seven surface variables directly to 15 outputs, the network first learns a compact representation of the underlying ocean state (fusing SST, SSS, SSH, currents, and winds) and then decodes that representation.
- *Note:* This is a learned latent mathematical representation, not a guaranteed physical ocean state.

C. DEPTH-CONDITIONED PROFILE DECODER

The decoder receives the **latent ocean embedding** AND a **learned depth embedding**, predicting temperature conditioned on depth. For each requested depth: $T(z) = \text{Decoder}(\text{latent ocean embedding}, \text{depth embedding})$

- **Why this is preferable:** It explicitly conditions prediction on depth, encourages shared information across the profile, makes the decoder naturally extensible to depth queries within the supported range, and gives the architecture a clear profile-reconstruction interpretation.
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13. End-to-End Methodology

STEP 1 — Data acquisition

Obtain 7 surface variables, GLORYS subsurface temperature, and ARGO observations for independent validation.

STEP 2 — Spatial harmonization

Define NIO region (5°N–30°N, 45°E–105°E). Convert/regrid all datasets to $0.25^\circ \times 0.25^\circ$. Ensure consistent longitude/latitude conventions.

STEP 3 — Temporal harmonization

Convert data to daily aligned samples. Handle timestamps consistently. Document interpolation/aggregation decisions.

STEP 4 — Quality control

Detect invalid values. Apply variable-specific QC. Preserve masks. Do NOT blindly fill missing values without recording validity.

STEP 5 — Target construction (GLORYS)

Create 15-depth GLORYS temperature targets. Because GLORYS native vertical levels do not exactly match SIH depths, we will retain the 36 native levels sufficient to cover 1000m and interpolate locally to the required 15 depths. *Never interpolate a surface-only temperature field down to 1000m.*

STEP 6 — Normalization

Fit normalization statistics ONLY on training data. Apply the same transformation to validation/test data.

STEP 7 — Dataset construction

Input: 7 surface variables + 7 masks. Target: 15-depth temperature profile.

STEP 8 — Baselines

Train climatology/persistence, linear/statistical, point-wise MLP, XGBoost (if feasible), plain CNN, and plain U-Net.

STEP 9 — Proposed model

Train Masked Multi-Scale U-Net → Latent Ocean Embedding → Depth-Conditioned Profile Decoder.

STEP 10 — Ablation

Test: U-Net without embedding bottleneck, U-Net + embedding, U-Net + depth conditioning, and full architecture. Every component must justify itself through measured improvement.

STEP 11 — Temporal extension

Only if spatial model leaves meaningful temporal information unused.

STEP 12 — Uncertainty extension

Only after deterministic reconstruction is stable.

STEP 13 — Independent ARGO validation

Match predictions to ARGO using spatial colocation tolerance, temporal colocation tolerance, and depth interpolation. Do NOT require exact grid/date matching. Evaluate RMSE, correlation, bias, and depth-wise performance.

14. Leakage Prevention & Scientific Guardrails

[Data Leakage]

- Use chronological splitting (e.g., Train: 2012-2020, Val: 2021, Test: 2022).
- Prevent spatial leakage (patch overlaps) and normalization leakage.
- ARGO must remain strictly independent of the training loop.

[Physics / Scientific Guardrails] We will **NOT** blindly impose: *"temperature must monotonically decrease with depth."* Temperature alone does not determine density stability (salinity affects density). Temperature inversions and barrier-layer effects can occur. Therefore, any physics loss should be introduced only after literature review and empirical validation. Avoid fake "physics-informed AI" claims.

15. MVP vs Advanced Version

Advanced components are NOT allowed to delay a validated MVP.

MVP Phase	Advanced Phase (Optional/Experimental)
NIO subset	3–7 day temporal context
7 surface variables + masks	Uncertainty estimation
15-depth GLORYS target	Lightweight attention / ConvGRU
0.25° daily grid	Improved missing-data robustness
Baseline models	Event/regime analysis
Masked U-Net + Latent Embedding	Explainability / feature attribution
Depth-Conditioned Decoder	
ARGO validation	

16. Experimental Ladder & Ablation Strategy

At every stage, we measure improvement. Do not assume the proposed model wins. Use Go/No-Go criteria.

```
graph TD
  A[Exp 0: Data pipeline validation] --> B[Exp 1: Climatology / persistence]
  B --> C[Exp 2: MLP]
  C --> D[Exp 3: XGBoost / linear]
  D --> E[Exp 4: CNN]
  E --> F[Exp 5: U-Net]
  F --> G[Exp 6: U-Net + Latent Ocean Embedding]
  G --> H[Exp 7: U-Net + Embedding + Depth Conditioning]
  H -.-> I[Exp 8: Temporal extension]
  I -.-> J[Exp 9: Uncertainty]
```

17. Feasibility Analysis

Can a 6-Student Team Actually Build This?

- **Data feasibility:** [GREEN] GLORYS is available. SIH requires a manageable NIO subset. We will start with a 7-day / 1-month pilot and scale only after successful audit.
- **Compute feasibility:** [GREEN] The 0.25° NIO grid is dramatically smaller than the full global native GLORYS grid. We can train on spatial patches/batches, use mixed precision, and start with a compact U-Net rather than a giant Transformer.
- **Engineering feasibility:** [YELLOW] Requires proficiency in PyTorch, xarray, netCDF, NumPy, and regridding tools. Reproducible preprocessing scripts are mandatory.
- **Team feasibility:** [GREEN] 6 members are sufficient if responsibilities are strictly divided.

18. Six-Member Team Execution Plan

Member	Responsibilities	Dependencies
Member 1	Data acquisition + preprocessing	Unblocks Member 2 & 3
Member 2	GLORYS target + vertical interpolation	Depends on Member 1
Member 3	Baselines (MLP, CNN)	Depends on Member 2
Member 4	U-Net + embedding + decoder	Depends on Member 3
Member 5	ARGO validation + metrics	Independent until inference
Member 6	Integration + experiments + documentation	Integrates 4 & 5

19. Timeline / Implementation Phases

- **Phase 1:** Data acquisition and pipeline validation.
- **Phase 2:** Baseline establishment.
- **Phase 3:** MVP Architecture (Masked U-Net + Embedding).
- **Phase 4:** Ablations & Advanced features (if time permits).
- **Phase 5:** ARGO validation & final metrics.
- **Phase 6:** Dashboard/Demo integration.

20. Go / No-Go Gates

- **GATE 1:** Can we obtain valid 3D GLORYS data covering 0–1000+ m? *(If NO: Do not proceed with fake/interpolated targets).*
- **GATE 2:** Can the data pipeline produce valid 0.25° daily NIO samples? *(If NO: Fix pipeline before modeling).*
- **GATE 3:** Does U-Net outperform point-wise baselines? *(If NO: Reconsider spatial architecture).*
- **GATE 4:** Does latent embedding improve reproducibly? *(If NO: Remove it).*
- **GATE 5:** Does depth conditioning improve profile reconstruction? *(If NO: Do not force it).*
- **GATE 6:** Does temporal modeling improve independent validation enough to justify complexity? *(If NO: Do not include it).*
- **GATE 7:** Does the final system improve against independent ARGO validation? *(If NO: Do not claim success).*

21. Judge-Ready Explanation

HOW WE WILL EXPLAIN OCEANEMBED TO THE JUDGES:

"Surface satellites observe the ocean skin, but the SIH problem asks us to reconstruct temperature hundreds of meters below it. Instead of treating every grid cell and depth independently, OceanEmbed first learns the spatial structure of the surface ocean and compresses the joint seven-variable state into a latent ocean representation. A depth-conditioned decoder then uses that representation to reconstruct the temperature profile at the required 15 depths. We compare this against simpler baselines and validate independently against ARGO, so every architectural component has to earn its place through measurable improvement."

22. Likely Judge Questions & Answers

- **Why U-Net?** It naturally handles dense spatial representations and captures mesoscale features (like eddies) at multiple scales.
- **Why not ConvLSTM?** Temporal evolution (ConvLSTM) adds massive compute and memory overhead. We test spatial context first; temporal modeling is an advanced extension evaluated only if spatial models leave information unused.
- **Why not Transformer?** ViTs are computationally expensive and data-hungry. A CNN/U-Net is a stronger starting point for a regional 0.25° grid.
- **Why use GLORYS?** It provides a continuous 3D historical record via NEMO reanalysis, making it the best available continuous target for ML training.
- **Why is ARGO independent?** ARGO floats provide real-world observational profiles. Validating against ARGO proves the model generalizes to reality, not just the GLORYS simulator.
- **Why masks?** Satellite data is often obscured by clouds. Masks allow the network to differentiate between a valid "zero" and missing data.
- **What is the Ocean Embedding?** A learned, compressed mathematical representation of the joint surface state (SST, SSS, SSH, currents, winds).
- **Why depth conditioning?** It explicitly tells the decoder which depth it is predicting for, acknowledging that the 15 outputs are physically connected points on a single vertical profile, not unrelated channels.
- **How do you prevent leakage?** Strict chronological splitting (e.g., train on 2012-2020, test on 2022). Normalization statistics are built only on training data.
- **What happens when satellite data are missing?** The model uses the validity mask to ignore missing pixels, relying on spatial context from surrounding valid pixels to infer the latent state.
- **How will you prove your model is better?** Through strict, component-wise ablation (MLP vs CNN vs U-Net vs OceanEmbed) and independent ARGO metrics (RMSE, correlation).
- **Is the model physically trustworthy?** We do not blindly force monotonic temperature rules (which fail for barrier layers). Trust is established strictly via ARGO validation.
- **What happens at 1000 m?** The model infers the deep structure through surface dynamic height (SSH) signatures and learned stratification correlations.
- **What is your MVP?** A spatial CNN/U-Net predicting the 15 depths on a real 0.25° dataset, validated against ARGO.
- **What is actually novel?** The combination of masked spatial encoding, learned latent ocean states, and depth-conditioned decoders, proven through strict ablation and independent validation.

23. Final Team Decision Sheet

Question	Answer
What is the simplest viable solution?	An MLP regression on point-wise surface data.
What obvious approaches are rejected?	Global GNNs, pure un-hybridized ViTs, "architecture soup".
Why are they rejected?	Unnecessary complexity for a regular 0.25° grid; compute constraints.
What is our strongest baseline?	A spatial CNN or plain U-Net.
What is our proposed advanced architecture?	Masked Multi-Scale U-Net + Latent Ocean Embedding + Depth-Conditioned Decoder.
Why this architecture?	It explicitly handles spatial context, joint variable representation, and profile-based decoding.
What is potentially novel?	Missing-data masking, depth-aware decoders, and uncertainty bounds (if proven).
What must be experimentally proven?	That the latent embedding and depth decoder actually improve ARGO metrics over a plain CNN.
What is the MVP?	Spatial CNN/U-Net, real 0.25° data, 15-depth target, ARGO validation.

What is the biggest feasibility risk?	Data acquisition pipelines and ARGO colocation scripting.
What happens if the advanced model fails?	We fall back to the highest-performing simple baseline (e.g., Plain CNN) that beats ARGO.

24. References

1. **[Official Source]** Smart India Hackathon 2026 Problem Statement 26066.
2. **[Official Source]** Copernicus Marine Service. *Global Ocean Physics Multiyear Product*. Dataset: `cmems_mod_glo_phy_my_0.083deg_P1D-m`.
3. **[Official Source]** GLORYS12V1 Reference. DOI: <https://doi.org/10.48670/moi-00021>
4. **[Official Source]** ARGO Data Management. Gridded ARGO / INCOIS LAS. *(Additional peer-reviewed literature citations pending formal literature review completion).*